Final Report. Building a Knowledge Graph for Chronic Disease Analysis

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Abstract

The growing availability of population health datasets offers new opportunities to analyse the relationships between chronic diseases and their associated risk factors. However, these data sources are typically distributed across independent tabular datasets, limiting the ability to capture complex interactions between patients, conditions and clinical indicators. This work proposes a Knowledge Graph (KG) framework to integrate heterogeneous public health data related to chronic diseases. Publicly available datasets describing diabetes, hypertension and stroke were collected and transformed into a semantic representation. The resulting KG contains nearly 30,000 patient entities and more than 100,000 clinical observations, providing a rich representation of health indicators and disease relationships. Query-based analysis of the graph reveals patterns of disease prevalence, common risk factors and comorbidity relationships within the population. These results demonstrate how KG technologies can facilitate the semantic integration of epidemiological datasets while enabling interpretable exploration of chronic disease interactions. The proposed framework illustrates the potential of graph-based representations for supporting transparent health data analysis and future research in explainable health analytics and clinical decision-support systems.

I. Introduction

Chronic diseases such as diabetes, hypertension, cardiovascular disease and stroke represent one of the most significant global health challenges of the 21st century. According to the World Health Organization (WHO) [1], chronic diseases account for approximately 74% of all deaths worldwide. Despite the large amount of clinical and epidemiological data available through public health surveys, electronic health records and research studies, this information remains fragmented across tabular datasets. Risk factors such as obesity, smoking, hypertension and high cholesterol are often analysed in isolation. Current data infrastructures typically rely on relational databases that do not explicitly represent complex relationships between risk factors, diseases and clinical measurements. Moreover, as healthcare systems increasingly rely on data-driven decision-making, there is a need for structured, semantically rich representations of medical knowledge that allow advanced querying, reasoning and integration of heterogeneous data sources.

A Knowledge Graph (KG) [2] offers a principled way to represent entities (e.g., diseases, risk factors, clinical measurements) and their relationships in a machine-interpretable format, enabling more transparent and flexible analysis than traditional approaches. The fragmentation of chronic disease knowledge negatively impacts patients, clinicians, public health authorities and healthcare systems by limiting early risk identification, underexploring comorbidity patterns and increasing preventable complications and costs. Doctors can utilise KGs for disease prediction and diagnosis, thereby improving medical outcomes and accuracy.

The remainder of the report is structured as followed. Firstly, state of the art is reviewed. Then, the methodology is presented, covering all steps from data loading and preprocessing to knowledge graph construction and query-based analysis. Section IV showcases the results obtained during the execution of the experiments. Finally, the report concludes with a discussion of the key findings, limitations and directions for future work.

II. Related Work

Knowledge Graphs (KGs) have become an important paradigm for representing and integrating heterogeneous biomedical data. Recent surveys highlight their growing role in healthcare analytics, enabling structured representation of clinical knowledge, improved data interoperability and advanced reasoning across complex biomedical entities [3]. By modeling entities such as diseases, symptoms, risk factors and measurements as interconnected nodes, KGs facilitate both knowledge discovery and explainable clinical analysis.

Early work by Bodenreider (2004) introduced the Unified Medical Language System (UMLS), which enables interoperability across multiple biomedical terminologies [4]. Similarly, SNOMED CT provides a comprehensive ontology of clinical concepts and relationships widely adopted in electronic health records and biomedical knowledge representation [5]. Ontology-based modeling has become a fundamental step in building interoperable biomedical knowledge graphs and integrating heterogeneous clinical data sources [6].

More recently, Chandak et al. (2022) introduced PrimeKG, a large-scale biomedical knowledge graph integrating diseases, drugs, genes, and phenotypes to support precision medicine and link prediction tasks [7]. Rotmensch et al. (2017) proposed embedding-based approaches over electronic health record graphs to improve diagnosis prediction, demonstrating that graph structures can enhance predictive modeling [8]. Zhang et al. (2020) developed a clinical KG framework for intelligent medical question answering and diagnosis support, showing improved interpretability compared to black-box machine learning systems [9]. Additionally, Li et al. (2021) presented a Chinese clinical Knowledge Graph for chronic disease diagnosis and treat-

ment, integrating hospital EHRs and treatment guidelines to improve multimorbidity management [10].

Although these works demonstrate the effectiveness of KGs in biomedical research, most existing approaches focus on molecular or pharmacological integration or rely on private EHRs. There is comparatively less work on building transparent, reusable KGs from publicly available epidemiological datasets to model shared behavioral and metabolic risk factors and comorbidity pathways in chronic diseases such as diabetes, hypertension and stroke. This project addresses that gap by constructing a KG from open-access population health datasets, explicitly modeling shared risk factors and enabling interpretable querying of disease interconnections.

III. Methodology

The process begins with defining a conceptual schema that captures core entities—patients, risk factors, clinical indicators and diseases, with their respective relationships. The conceptual structure of the model is illustrated in the entity diagram provided in the Appendix. Patients represent the central entities in the graph and are connected to diseases, risk factors and clinical indicators through semantic relationships. The KG was implemented using the RDF data model with the *rdflib* library in Python.

The public datasets used in this study were obtained from the Health Dataset on Kaggle [11], which contains population-level information on diabetes, hypertension, stroke and associated health indicators (e.g., high blood pressure, glucose level), as well as lifestyle factors (e.g., smoking status, physical activity). Worth mentioning, the full framework, including scripts and outputs, is available on GitHub[1]. The workflow follows these steps: data preprocessing, schema definition, knowledge graph construction and execution of queries to validate and explore the graph.

During the data loading and cleaning process, special attention was given to handling missing values, ensuring unique patient identifiers, and verifying that all numerical and categorical measurements were correctly typed, with unique patient identifiers. To maintain data consistency, rows containing missing values were removed. Since the amount of missing values is relatively small, deleting these rows does not significantly affect the dataset size. This approach guarantees a coherent and queryable representation. Correlation matrices were computed for the main target variables (diabetes, hypertension and stroke) to explore relationships among health indicators and to guide feature selection for knowledge graph construction. For instance, in the diabetes dataset, Stroke, HeartDiseaseorAttack and DiffWalk showed the highest positive correlations (see Figure 1). Similarly, correlation matrices for the hypertension and stroke datasets, are provided in the Appendix.

After defining the conceptual schema, the cleaned datasets were transformed into RDF triples, formalising

[1]Code available at: `https://github.com/andreapradas/KG_Final_Project`

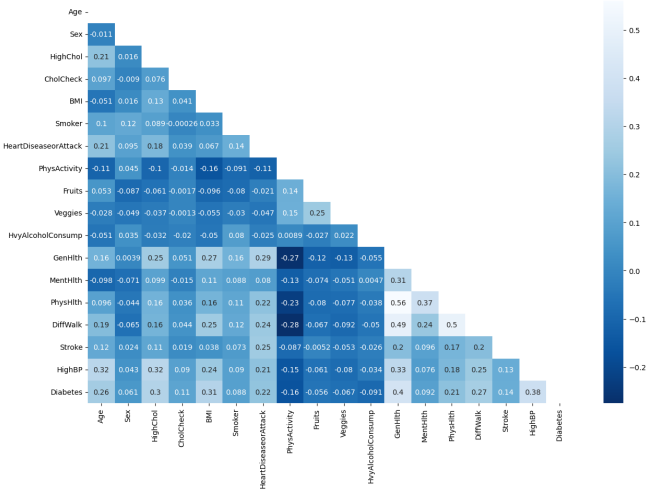

Fig. 1: Correlation matrix for the diabetes dataset. Strong positive correlations can be observed for Stroke, HighBP and DiffWalk.

entities and their relationships according to the schema. Each CSV file was mapped to nodes and edges, with patients represented as individual nodes connected to disease nodes, risk factor nodes and measurement nodes through semantic relationships (`hasDisease`, `hasRiskFactor`, `hasMeasurement`). Risk factors such as smoking, high cholesterol and high blood pressure are represented as instances of the class `schema:MedicalRiskFactor`. Clinical measurements were represented as instances of `schema:MedicalObservation`, each observation stores the measurement name and value and is linked to the corresponding patient entity. Wherever possible, standardised medical vocabularies such as SNOMED CT were used to represent disease concepts, ensuring semantic interoperability. The resulting knowledge graph was then serialised to a Turtle file as `chronic_diseases_KG.ttl`, making it reusable for subsequent analyses. The file is included in the output folder of the project zip file.

Once the Knowledge Graph was constructed, SPARQL queries [12] were used to evaluate to its quality and extract actionable insights. The first set of queries focused on data quality assessment, verifying completeness, correctness, consistency and coherence of the graph. For instance, queries counted missing relationships, checked data types and unique identifiers, validated that relationships conformed to expected domains and ranges and detected orphan nodes or cyclic inconsistencies. This process ensures that the KG provides a reliable and semantically consistent representation of the population health data.

Beyond quality assessment, additional queries were designed to explore the graph and reveal hidden patterns in the data. These queries enabled the analysis of disease prevalence, identification of common risk factors, exploration of comorbidity patterns and detection of patients with multiple risk factors but no diagnosed disease. By supporting interpretable, query-driven exploration of relationships between

patients, diseases, risk factors and clinical measurements, the KG demonstrates its practical utility for both descriptive analysis and hypothesis-driven investigation.

The main contribution of this work lies in enhancing semantic interoperability across heterogeneous health datasets while providing a reusable technical framework. By constructing a knowledge graph that explicitly models patients, diseases, risk factors and clinical measurements, the pipeline enables interpretable SPARQL-based querying, systematic data quality assessment and potential integration with external biomedical knowledge graphs. This approach not only standardizes variable definitions across datasets but also facilitates exploration of disease interconnections and shared risk factors.

IV. Results

This section presents the outputs obtained from the construction and analysis of the KG. The results include the structural characteristics of the graph as well as the outputs of the SPARQL queries used to evaluate its content and analytical capabilities. The generated knowledge graph integrates patient data, diseases, risk factors and clinical measurements into a unified semantic structure.

Figure 2 illustrates a subgraph extracted from the KG, showing two example patient nodes together with their associated diseases, risk factors and measurements. This visualisation highlights the semantic structure of the graph and the relationships between different health entities.

To evaluate the KG, several SPARQL queries were executed. The results are organised into two categories: (i) data quality assessment queries used to verify the integrity of the graph and (ii) analytical queries designed to extract insights about disease prevalence, risk factors and comorbidity patterns. The structural composition of the KG is summarised in Table I, which reports the number of instances for the main classes represented in the graph.

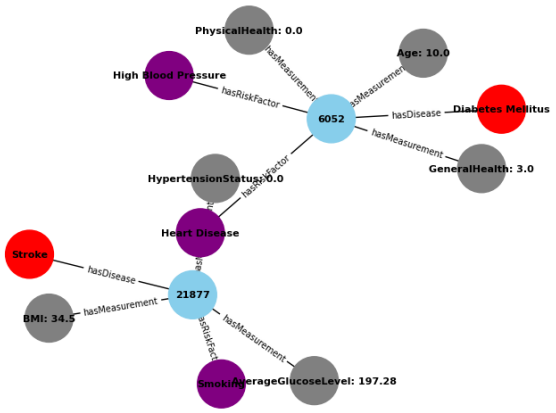

Fig. 2: Shows a subgraph of two random patients with ids 21877 and 6052 (blue nodes: patients, red nodes: diseases, purple nodes: risk factors and grey nodes: measurements).

TABLE I: Counts of main classes in the knowledge graph.

Class	Count
Disease	3
MedicalRiskFactor	6
Patient	29,990
MedicalObservation	119,940

Another query was executed to identify potential missing relationships within the graph. Table II reports the number of patients that are not linked to any risk factor.

TABLE II: Number of patients without any associated risk factor.

Description	Count
Patients with no risk factor	11,758

The following queries focus on extracting descriptive statistics from the KG. Table III presents the number of patients associated with each chronic disease represented in the graph.

TABLE III: Results for the query: prevalence of the main chronic diseases in the hole dataset.

Disease	#Patients
Diabetes Mellitus (SNOMED: 44054006)	10,631
Hypertension (SNOMED: 38341003)	5,494
Stroke (SNOMED: 230690007)	4,961

The distribution of risk factors across the entire patient population is summarised in Table IV.

TABLE IV: Top risk factors across all patients.

Risk Factor	#Patients
High blood pressure	8,814
High cholesterol	5,322
Smoking	5,257
Heart disease	3,517
Exercise-induced angina	3,271
Difficulty walking	2,579

Additional queries explored the distribution of risk factors within specific diseases. The results for diabetes, hypertension and stroke are reported in Tables V, VI and VII, respectively.

TABLE V: Distribution of risk factors among patients diagnosed with diabetes (SNOMED: 44054006).

Risk Factor	#Patients
High blood pressure	3,730
High cholesterol	3,391
Difficulty walking	1,866
Heart disease	1,126

TABLE VI: Risk factors associated with patients diagnosed with hypertension (SNOMED: 38341003).

Risk Factor	#Patients
Exercise-induced angina	759

TABLE VII: Distribution of risk factors among patients diagnosed with stroke (SNOMED: 230690007).

Risk Factor	#Patients
Smoking	5,257
High blood pressure	3,693
Heart disease	2,350
High cholesterol	469
Difficulty walking	374

Queries were also executed to identify examples of comorbidity and patients presenting multiple risk factors. A sample of patients diagnosed with more than one disease is presented in Table VIII, while Table IX reports patients with the highest number of associated risk factors.

TABLE VIII: Sample of patients with multiple chronic diseases.

Patient	Disease 1	Disease 2
29	Diabetes	Stroke
40	Diabetes	Stroke
41	Diabetes	Stroke
48	Diabetes	Stroke
51	Diabetes	Stroke
⋮	⋮	⋮
279	Diabetes	Stroke

TABLE IX: Top patients with the highest number of risk factors.

Patient	#Risk Factors
Patient 12	4
Patient 19	4
Patient 45	4
Patient 60	4
Patient 61	4
⋮	⋮
Patient 275	4

V. Discussion

This section discusses the main findings derived from the KG analysis and interprets the results presented in the previous section. The objective is to examine the observed disease prevalence, risk factor distributions and comorbidity patterns, also to understand their implications for chronic disease modeling.

First of all, the data quality assessment confirmed that all diseases, risk factors and clinical measurements were correctly typed and semantically consistent, ensuring that the KG is coherent, complete and ready for downstream analytical queries. This verification was performed through a set of validation queries, whose outputs are presented in Tables I and II.

The prevalence results obtained from the KG provide an overview of the distribution of chronic diseases within the dataset (see Table III). Stroke appears as the most frequent condition, affecting 10,631 patients, while hypertension and diabetes are present in 5,494 and 4,961 patients respectively. This distribution offers an initial picture of the health profiles represented in the data and establishes a baseline for further analyses, particularly in the context of comorbidity patterns and patient risk stratification.

When examining the overall distribution of risk factors across the patient population, high blood pressure emerges as the most common condition, affecting 8,814 individuals. High cholesterol and smoking are also widely present, with 5,322 and 5,257 patients respectively. Other factors such as heart disease, exercise-induced angina and difficulty walking appear less frequently but remain relevant within the population. Together, these results, presented in Table IV, highlight the strong presence of cardiovascular and lifestyle-related risks within the dataset.

Looking more closely at the relationship between diseases and their associated risk factors reveals different patterns for each condition. Among patients diagnosed with diabetes (shown in Table V), high blood pressure and high cholesterol are the most frequently observed factors, followed by difficulty walking and heart disease. In the case of hypertension (Table VI), exercise-induced angina shows a notable presence. For patients suffering from stroke (Table VII), smoking and high blood pressure appear most frequently, with heart disease, high cholesterol and difficulty walking also present. These observations suggest that different diseases exhibit distinct combinations of risk factors within the population.

The analysis of comorbidities further indicates that several patients present more than one chronic condition simultaneously. In particular, multiple cases show the co-occurrence of diabetes and stroke within the same patient profiles. Such patterns highlight the complexity of chronic disease interactions and underline the importance of considering comorbidity when analysing patient health trajectories.

Finally, examining the number of risk factors associated with each patient reveals that many individuals present multiple concurrent risks. Several patients in the dataset exhibit up to four risk factors simultaneously, illustrating the multifactorial nature of chronic disease development. This distribution reflects the cumulative burden of risk factors within the population and provides useful context for understanding the overall health profile captured in the knowledge graph.

VI. Conclusion

In conclusion, this project demonstrates how KG technologies can be used to integrate heterogeneous health datasets and represent relationships between patients, diseases, risk factors and clinical indicators in a unified semantic structure. The resulting KG enables structured exploration of disease prevalence, risk factor distributions and comorbidity patterns through interpretable queries, illustrating the potential of graph-based approaches for analysing complex health

data.

Despite these contributions, several limitations should be acknowledged. The graph is constructed from publicly available datasets that may contain limited clinical detail and simplified variables compared to real-world electronic health records. In addition, the current implementation focuses on descriptive analysis rather than predictive modeling.

Future work could extend the graph by incorporating additional biomedical datasets, integrating richer clinical ontologies and applying graph-based machine learning techniques for predictive risk modeling. Such developments could further support explainable AI applications and clinical decision-support systems built upon semantically structured health data.

REFERENCES

[1] World Health Organization. Noncommunicable diseases, 2023.

[2] Aidan Hogan, Erik Blomqvist, Michael Cochez, Claudia d'Amato, Georgiana de Melo, Carlos Gutierrez, Sean Kirrane, Roberto Navigli, Sebastian Neumaier, Axel Polleres, and et al. Knowledge graphs. *ACM Computing Surveys*, 2021.

[3] Bilal Abu-Salih, Muhammad AL-Qurishi, Mohammed Alweshah, Mohammad AL-Smadi, Reem Alfayez, and Heba Saadeh. Healthcare knowledge graph construction: A systematic review of the state-of-the-art, open issues, and opportunities. *Journal of Big Data*, 2023.

[4] Olivier Bodenreider. The unified medical language system (umls): integrating biomedical terminology. *Nucleic Acids Research*, 2004.

[5] SNOMED International. Snomed ct, 2023. Accessed: 2026-02-27.

[6] Natalja Kurbatova and Rowan Swiers. Disease ontologies for knowledge graphs. *BMC Bioinformatics*, 2021.

[7] Pritha Chandak, Kun Huang, and Marinka Zitnik. Building a knowledge graph to enable precision medicine. *Scientific Data*, 2023.

[8] Halpern Y. Tlimat A. Horng S. Sontag D. Rotmensch, M. Learning a health knowledge graph from electronic medical records. *Scientific Reports*, 2017.

[9] Tang B. Wang X. Xu H. Zhang, Y. A clinical knowledge graph for intelligent question answering and diagnosis support. *Journal of Biomedical Informatics*, 2020.

[10] X. Li, Y. Wang, and Z. Chen. A clinical knowledge graph for chronic disease diagnosis and treatment in chinese hospitals. *Journal of Biomedical Informatics*, 2021.

[11] Prosper Chuks. Health dataset (diabetes, hypertension and stroke prediction). `https://www.kaggle.com/datasets/prosperchuks/health-dataset`, 2024.

[12] Carlos Buil-Aranda, Marcelo Arenas, Oscar Corcho, and Axel Polleres. Federating queries in sparql 1.1: Syntax, semantics and evaluation. *Web Semant.*, 2013.

APPENDIX

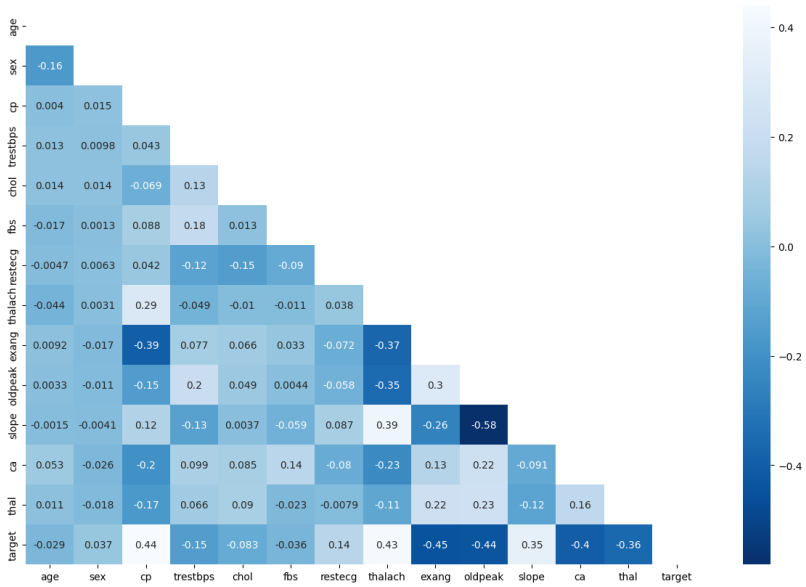

Fig. 4: Correlation matrix for the hypertension dataset.

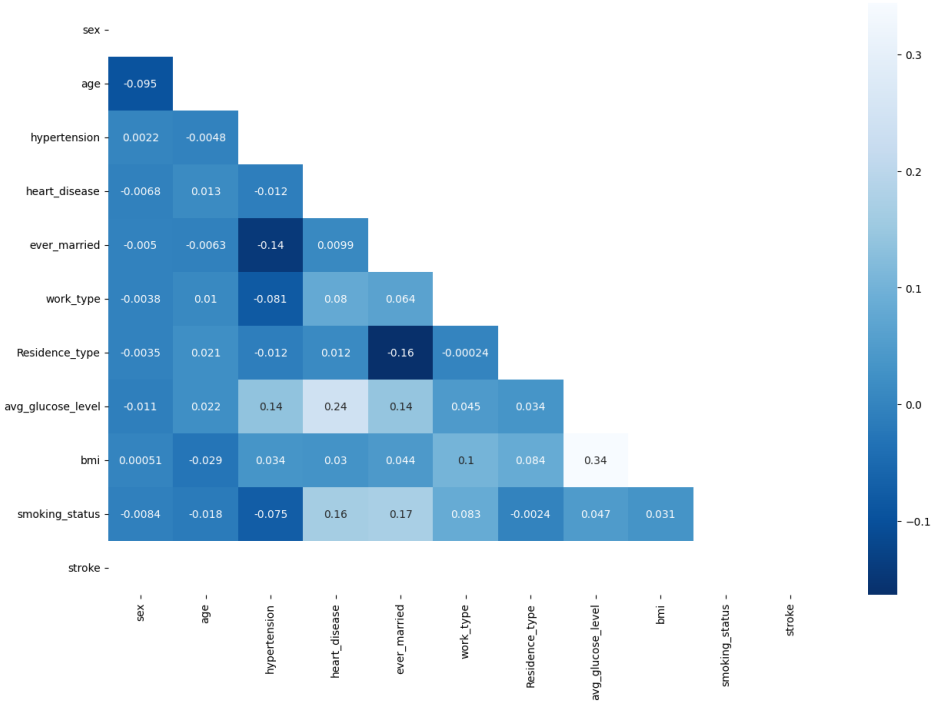

Fig. 5: Correlation matrix for the stroke dataset.

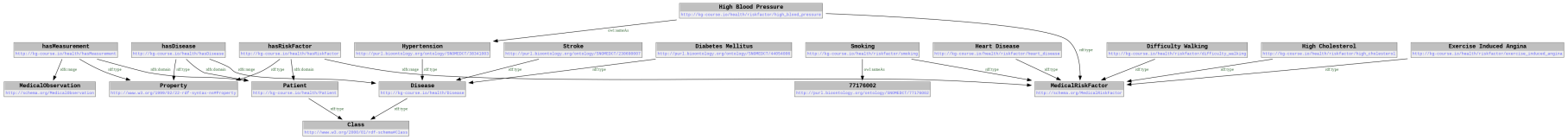

Fig. 3: Schema of the knowledge graph, illustrating entities (patients, diseases, risk factors, measurements) with their relationships.